

Department of Nano Science Technology

Faculty of Technology

Wayamba University of Sri Lanka

Centralized Workshops and Laboratory Allocation Platform

UI Navigation and Screen Flow

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1. Introduction – Purpose of the Navigation Flow

The primary objective of this Navigation and Screen Flow Specification is to establish an intuitive, unambiguous, and error-resilient interaction model for the Campus Facility Allocation System. By explicitly mapping each screen transition, decision point, conditional branch, and security boundary, this document ensures that developers, UI designers, and academic stakeholders maintain a shared vision of the user journey from entry to execution.

Core UX Design Tenets

• Single Responsibility Principle: Every GUI view focuses on one operational milestone to eliminate cognitive overload.
• Predictable State Progression: Linear forward momentum backed by persistent 'Back' and 'Cancel' escape hatches.
• Zero Silent Failures: Every input validation failure or scheduling conflict yields immediate inline contextual guidance.
• Cryptographic Accountability: Reservations are secured and identified by unique 8-character tokenized keys.

Target User Roles & Interaction Scope

- **Student Applicant:** Conducts self-service facility availability checks, reserves slots, and receives cryptographic booking keys.
- **Reservation Holder (Student):** Accesses existing records via 8-character alphanumeric allocation keys to modify schedules or cancel reservations.
- **Facility Administrator:** Monitors real-time venue utilization, conducts manual scheduling overrides, and manages conflict resolution.
- **System Administrator (Root):** Maintains platform configurations, themes, wallpapers, and administrator credential security via Master Passcode.

2. Overall Screen Flow Diagram – Complete System Navigation

The Overall Screen Flow represents the high-level operational topography of the entire facility management application. The Welcome GUI serves as the single centralized dispatch hub. Depending on the user persona and intent, the interaction diverges into four strictly encapsulated subsystems: Student Booking, Reservation Modification, Administrative Oversight, and Root System Configuration.

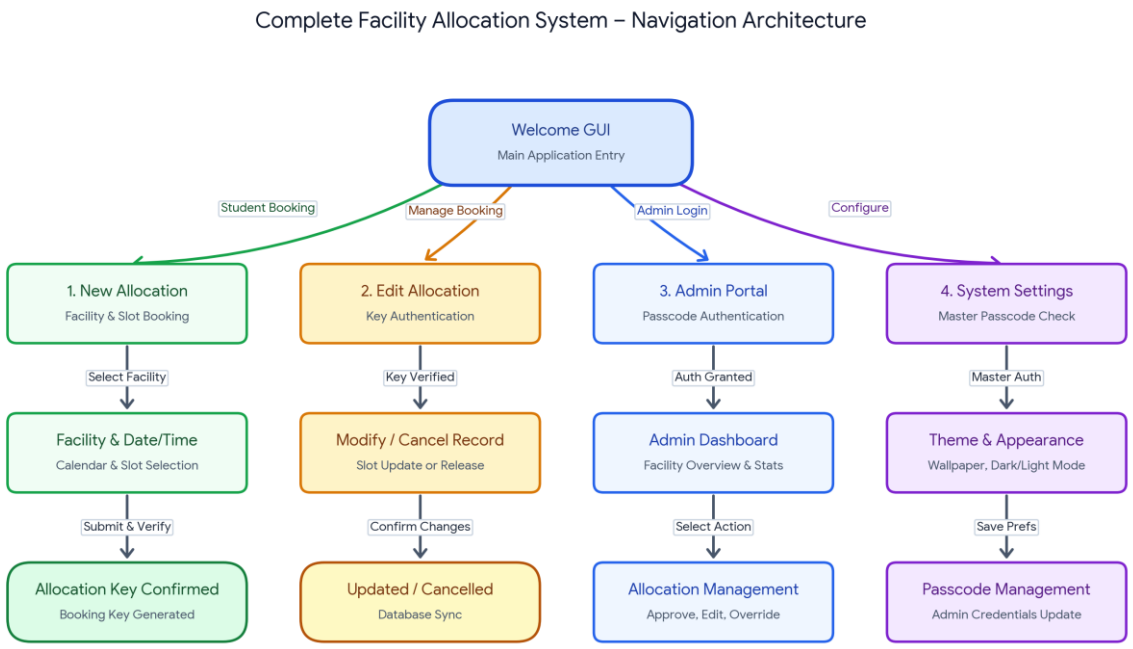


Figure 2.1: High-Level Navigation Architecture and System Module Dispatch Flow

Global Subsystem Dispatch Matrix

Subsystem Route	Target Persona	Entry Trigger / Action	Security & Validation Boundary
1. Student New Allocation	General Students	Select 'New Allocation' button on Welcome GUI	Open public access; client-side conflict check
2. Edit Allocation Flow	Existing Booking Holders	Select 'Edit Allocation' button on Welcome GUI	Requires 8-character cryptographic Allocation Key
3. Administrator Portal	Facility Managers	Select 'Admin Login' & select Target Facility	Requires facility-specific hashed Administrator Passcode
4. System Settings	Lead SysAdmin / SuperUser	Select 'System Settings' cog icon	Master Passcode mandatory; elevated root authorization

3. Student Navigation Flow – New Facility Booking

The Student Navigation Flow is structured as a friction-free, 6-stage sequential wizard. It guides the student from the initial landing portal through facility browsing, availability slot selection, user profile registration, and final cryptographic key generation. A strict conflict-detection engine prevents double-booking before submission.

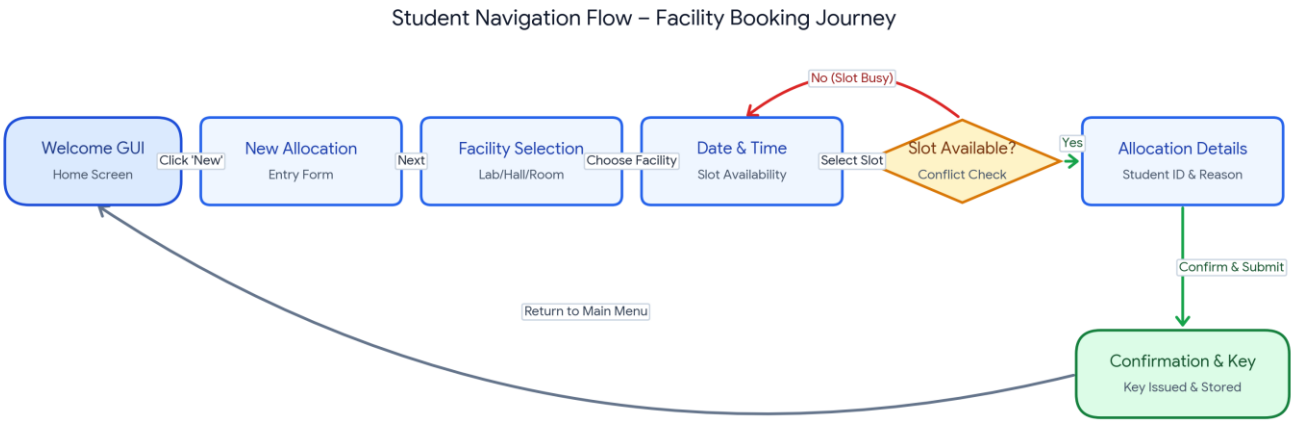


Figure 3.1: Student Multi-Step Facility Allocation Navigation Workflow

Detailed Screen Breakdown & Interactive Elements

- **3.1 Welcome GUI:** The clean, academic entry point presenting primary navigation tiles. Features real-time operational announcements and distinct calls to action.
- **3.2 New Allocation:** Initializes the reservation wizard, outlining eligibility guidelines, standard booking windows (up to 14 days in advance), and usage policies.
- **3.3 Facility Selection:** Presents a filterable gallery of campus resources (e.g., Computer Labs, Lecture Theatres, Collaborative Study Rooms) showing capacity and equipment.
- **3.4 Date & Time Selection:** Interactive date picker coupled with hourly time-block grids. Dynamically checks real-time database locks to highlight free vs. occupied slots.
- **3.5 Allocation Details:** Form capturing applicant credentials (Student ID, Full Name, Department, Academic Purpose, Supervisor Endorsement if required).
- **3.6 Confirmation & Allocation Key:** Terminal confirmation displaying reservation summary and generating a tamper-proof 8-character token (e.g., 'ALLOC-9B4X').

Student Navigation Transition Matrix

Stage	Screen Name	User Action	Validation / Business Logic	Destination Screen
1	Welcome GUI	Click 'New Allocation'	Check system online status	New Allocation Entry
2	New Allocation Entry	Click 'Begin Reservation'	Verify student session / term validity	Facility Selection
3	Facility Selection	Select venue & click 'Next'	Verify facility active & under quota	Date & Time Selection
4	Date & Time Selection	Pick date & slot interval	Real-time conflict check (Available: Pass / Busy: Alert)	Allocation Details
5	Allocation Details	Submit applicant details	Regex validation on ID, mandatory fields populated	Confirmation Screen
6	Confirmation Screen	Save key & return to Home	Commit transaction, generate key, release UI state	Welcome GUI

4. Edit Allocation Flow – Key Authentication, Modify & Cancel

To eliminate the overhead of student account passwords, reservations are managed via a zero-trust tokenized model. The Edit Allocation Flow empowers students to retrieve their existing booking using their Allocation Key. Once verified, users can either reschedule to an alternate slot or completely release the facility back to the general pool.

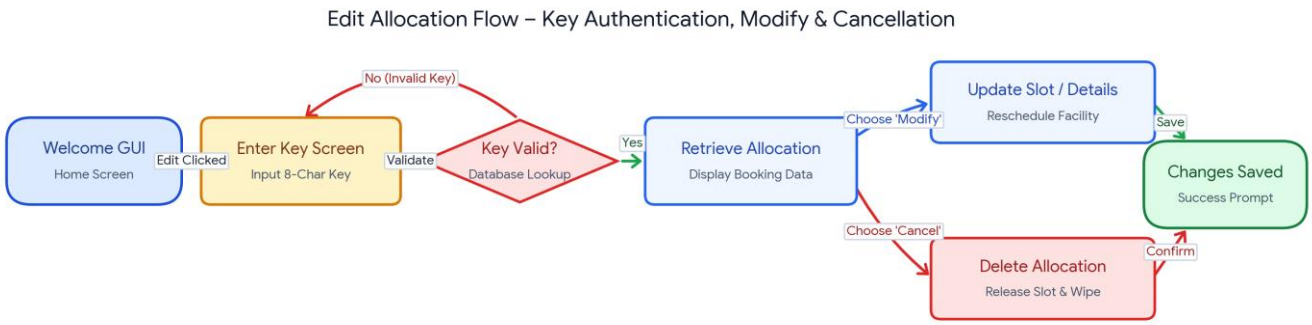


Figure 4.1: Allocation Key Verification, Schedule Rescheduling, and Cancellation Flow

Detailed Screen Breakdown & Interactive Elements

- **4.1 Allocation Key Authentication:** Secure input prompt equipped with 8-character masking and formatted entry. Includes brute-force lockout (maximum 5 attempts per IP).
- **4.2 Retrieve Existing Allocation:** Queries the central database and populates a read-only overview showing Facility Name, Scheduled Date, Time Slot, and Booking Timestamp.
- **4.3 Update / Delete Action Interface:** Two dedicated action pathways: 'Modify Slot' (opens date/time selection while holding existing reservation lock) or 'Cancel Allocation' (opens two-step confirmation modal).

Edit & Cancellation State Logic

User Intent	Screen Trigger	Condition & System Behavior	Final State / Feedback
Invalid Key Entered	Key Authentication Screen	Key not found in database or expired	Inline error display; remain on key prompt
Modify Slot	Active Allocation Overview	User picks new valid slot; system atomically swaps slots	Key retained; modified schedule confirmed
Delete / Cancel	Active Allocation Overview	User confirms release dialog; record purged from active pool	Slot returned to available pool; key invalidated

5. Administrator Navigation Flow – Facility Control & Audit

The Administrator Navigation Flow equips department coordinators and lab managers with facility-scoped governance tools. To maintain decentralized security, administrators first select their managed facility, supply their specific passcode, and access a tailored dashboard with comprehensive allocation monitoring and override authority.

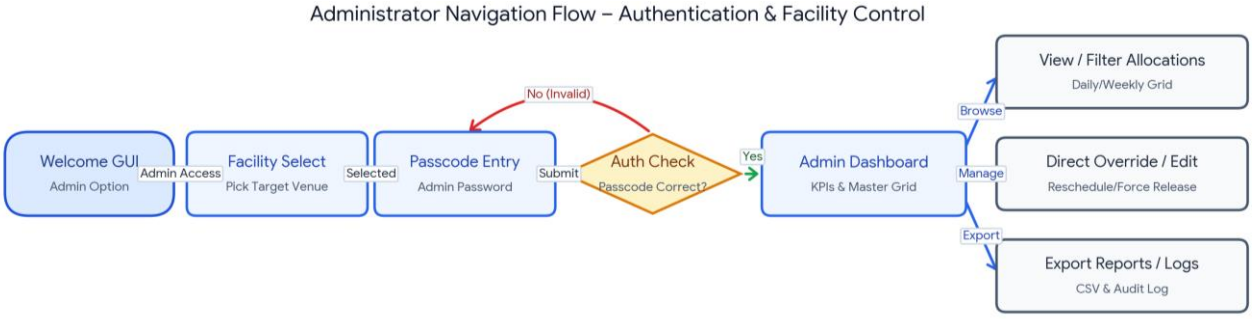


Figure 5.1: Administrator Authentication, Facility Dashboard, and Allocation Management

Detailed Screen Breakdown & Interactive Elements

- **5.1 Facility Selection:** Dropdown selector allowing the administrator to choose the venue they oversee (e.g., Robotics Lab, Auditorium, Seminar Hall A).
- **5.2 Administrator Passcode:** Facility-bound credential challenge screen. Validates hashed credentials with rate-limiting to prevent unauthorized management entry.
- **5.3 Administrator Dashboard:** Central command center displaying real-time occupancy metrics, daily/weekly timetables, and notification feeds for upcoming bookings.
- **5.4 Allocation Management:** Interactive data grid providing search, multi-parameter filtering, manual booking insertion, emergency cancellation, and audit log generation.

Administrator Operations & Control Matrix

Operational Function	Dashboard Component	Administrative Action	System Output
Facility Switch / Login	Facility Select & Passcode	Authenticate using assigned venue credentials	Grants role-scoped session token
Timetable Monitoring	Master Schedule Grid	Filter by Date, Slot, or Student ID	Displays current and upcoming occupancy
Administrative Override	Allocation Management	Force-reschedule or revoke conflicting bookings	Direct DB update; logs reason for audit
Reporting & Export	Export Controls	Click 'Generate Audit Log / CSV'	Generates downloadable booking report

6. System Settings Flow – Security & Appearance Configuration

The System Settings Flow represents the highest security tier within the application. Reserved for Lead Systems Administrators, this subsystem governs visual identity (themes, wallpapers) and administrative security policies. Entry is guarded by a cryptographically enforced Master Passcode.

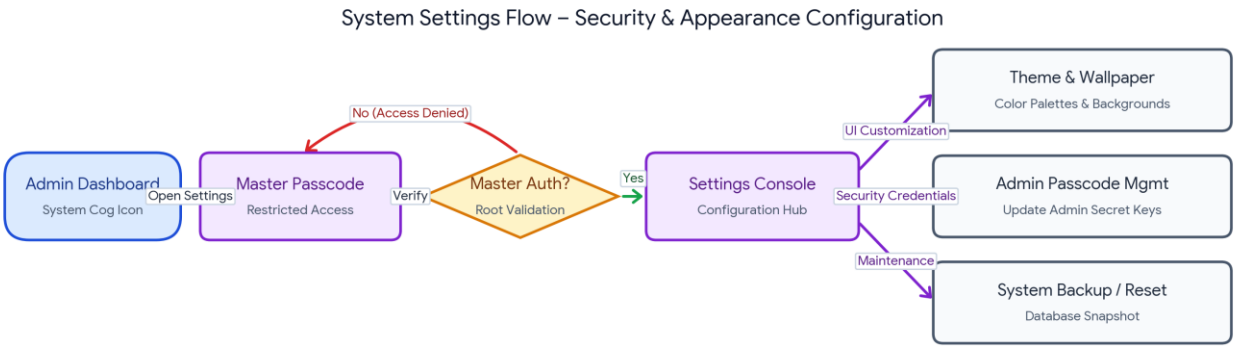


Figure 6.1: Master Passcode Authentication and System Configuration Workflow

Detailed Screen Breakdown & Interactive Elements

- **6.1 Master Passcode Authentication:** Strict root barrier prompt requiring the master administrative credential. Prevents casual or unauthorized setting changes.
- **6.2 Theme and Wallpaper:** Customization suite allowing institutions to set official branding colors (Navy, Slate, Emerald), dark/light high-contrast modes, and background wallpapers.
- **6.3 Administrator Passcode Management:** Central credential management console allowing the rotation and renewal of individual facility administrator passcodes without database tampering.

System Configuration & Governance Rules

Setting Category	Target Configuration	Permitted Modifications	Persistence Mechanism
Root Authentication	Master Passcode Validation	Re-authenticate; modify master root password	Salted hash in secure storage config
UI Customization	Theme & Wallpaper Engine	Select Light/Dark mode, primary theme, custom campus background	Saved to local JSON preferences file
Security Management	Facility Admin Passcodes	Reset/rotate individual lab and hall admin credentials	Encrypted hash update across facility table